

TinyC

Compiler Design Lab
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TinyC features

1. Procedural language
2. Statically typed
3. Set of operators : { Arithmetic operators, relational operators, ternary operator (?:), type casting operator }
4. Set of control flow statements: { if-else, while }
5. Set of data types : { int, double, char }
6. Single line comment
7. Limited I/O support : only output support(print)

TinyC program structure

1

Every TinyC program has an optional global declarations followed by one or more functions and one function must be main.

2

Every TinyC function has zero or more TinyC statements

3

Every TinyC statement is any of the following, an assignment statement, a control flow statement, a return statement, a declaration statement, a TinyC print statement

TinyC examples: Example 1

```
int main(){  
    int a,b,c;  
  
    a=5;  
  
    b=5;  
  
    c=a+b; // all statements are similar to C, except print  
  
    print c; // print is predefined operator in TinyC which prints the value of c  
  
}
```

TinyC example Example2

```
int a,b=40;
```

```
int main(){
```

```
    int a=30;
```

```
    c=a+b*a+a*a;
```

```
    print a;
```

```
}
```

TinyC examples: Example 3

```
int a=40,b=50;  
  
double c;  
  
int main(){  
    if(a>b)  
        c=(double)a/b;  
    else  
        c=(double)b/a;  
    print c;  
}
```

TinyC examples: Example4

```
int a,b,c;
double d,e,f;;
int add(int,int); //declaration of function
double mult(double,double);
int main(){
    c=add(30,40);
    d=multiply(40.8,50.2);
    a=-5;
    b=-45;
    c= add(a,b);
    print c;
}
Int add(int a1, int a2){
return a+b;
}
```

Implementation of TinyC

1. Write a compiler which takes TinyC program as an input and converts it to a valid assembly language code.
2. SPIM is used to test the generated assembly language code

Compilation:

Lexical analysis -> Syntax Analysis -> Semantic Analysis -> Intermediate code generation -> Code Optimization -> Code generation

Interpretation:

Lexical analysis -> Syntax Analysis -> Semantic Analysis -> Evaluate statements

Translation vs Evaluation: Example

```
int main(){  
    int a,b,c;  
  
    a=5;  
  
    b=25;  
  
    c=a*a+b-20; // all statements are similar to C, except print  
  
    print c; // print is predefined operator in TinyC which prints the value of c  
  
}
```

Evaluation gives :